Chapter 6

Probability Distributions

In order to associated more complex probabilities to a sample space, we need to employ probability distributions, which are models for uncertain outcomes. They allow us to move from statements such as

> *"Flights are sometimes late."*

to questions such as

> *"If 10% of flights are delayed, what is the probability that at least two of the next ten flights are delayed?"*

In a first part, the concept of random variables is presented including its expected value and variance. The two remaining parts of this chapter cover probability distributions for discrete and continuous random variables. The goal is to learn how to recognize a random process, choose a appropriate probability distribution, calculate probabilities in R, simulate uncertain outcomes, and interpret what the results mean.

Many probability distributions in R have a specific name associated with them, which is used in multiple functions. For discrete probability distributions such as the binomial or the Poisson distribution, `binom` and `pois` are used, respectively. The names `unif`, `norm`, and `t` are used for the continuous uniform, normal, and Student distributions, respectively. Note that the Student distribution is also known as the t-distribution. There is a particular set of R commands to find various aspects of the distributions:

- `dname()` for the density or probability function
- `pname()` for the cumulative density function
- `rname()` for the random numbers
- `qname()` for the inverse

6.1 Random Variables

Suppose a city operates a service center and in a typical hour, the number of incoming calls varies. Some hours have only a few calls while others have many. The exact number of calls during the next hour is unknown and the following questions may arise:

- How many calls should we expect?
- What is the probability of more than 15 calls?
- How frequently should unusually busy periods occur?
- How much staffing is needed if we want enough capacity 95% of the time?

These are different questions about the same uncertain process. A probability distribution describes which values an uncertain quantity can take and how likely those values are. The uncertain quantity is called a random variable. In the service center case, we can define the random variable X, which is the number of calls received during the next hour. The value of X is not known before the hour occurs. A probability distribution gives us a model for the possible values of X. A probability distribution is a model and not data, which contains observations that already occurred. For example, we might observe the following number of calls during ten hours: 8, 12, 9, 11, 15, 10, 7, 13, 9, and 14. If we decide that hourly calls can reasonably be represented by a count distribution with an average of 10.8 calls per hour, the Poisson distribution may be appropriate.

Random variables can be either discrete or continuous. A discrete random variable takes separate, countable values. A continuous random variable can take any value within an interval. Waiting time, height, temperature, income, and distance are often treated as continuous. This distinction matters because different probability distributions are appropriate for different kinds of variables. The important aspect to keep in mind is that the sum of the probabilities for all possible values of the random variable has to be equal to one.

Before presenting discrete and continuous distributions, let us first introduce the expected value and variance, which describe the center and spread of a random variable given its probability distribution. Think of the expected value as a weighted average with the probabilities being the weights. The expected value is the long-run average outcome of a random process.

If X is a discrete random variable taking the values $x_1, x_2, \ldots, x_n$ and with probability mass function P. Then the expected value of X is defined as follows

$$E(X) = \sum_i x_i \cdot \Pr(X = x_i)$$

Suppose you roll a die and observe the number that comes up. The probability mass or frequency function is given by

$$\Pr(X = x_i) = \frac{1}{6} \quad \text{for i=1,...,6}$$

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
Pr	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
X	3	2	2	1	2	1	1	0

Table 6.1: Number of heads as a random variable after three coin flips.

Thus, the expected value can be calculated as follows:

$$E(X) = \sum_{i=1}^{6} x_i \cdot \left(\frac{1}{6}\right) = 21/6 = 3.5$$

This does not mean that we expect to roll a 3.5 since a die cannot produce 3.5 but that repeating the experiment many times, the average approaches 3.5.

Let X be a continuous random variable taking the values with probability density function $f(x)$. Then the expected value of X is defined as follows

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)dx$$

While the expected value describes the center of a random variable, the variance measures how far outcomes tend to be from the expected value. The variance can be calculated in with to different equations:

$$Var(X) = E[(X - E(X))^2] = E(X^2) - E(X)^2$$

Both equations give you the variance with one of the equations being sometimes more convenient than the other. Keep in mind that $E(X^2) \neq E(X)^2$. In applied work, the standard deviation, which is the square root of the variance, is usually easier to interpret since it is expressed in the same units as the random variable.

6.2 Discrete Probability Distributions

A random variable X is said to be discrete if it can assume only a finite or countable infinite number of distinct values. A discrete random variable can be defined on both a countable or uncountable sample space. The probability that X takes on the value k, i.e., $\Pr(X = k)$, is defined as the sum of the probabilities of all sample points that are assigned the value k. For each value within its domain, we have $\Pr(X = k) \geq 0$ and that the sum of all probabilities is equal to one. Suppose you are flipping a coin three times and associate the outcome heads with the number one (Table 6.1).

This experiment can also be represented with a histogram which can be turned into a frequency histogram. This is illustrated for flipping a coin ten times and repeating the experiment 10,000 times (Figure 6.1).

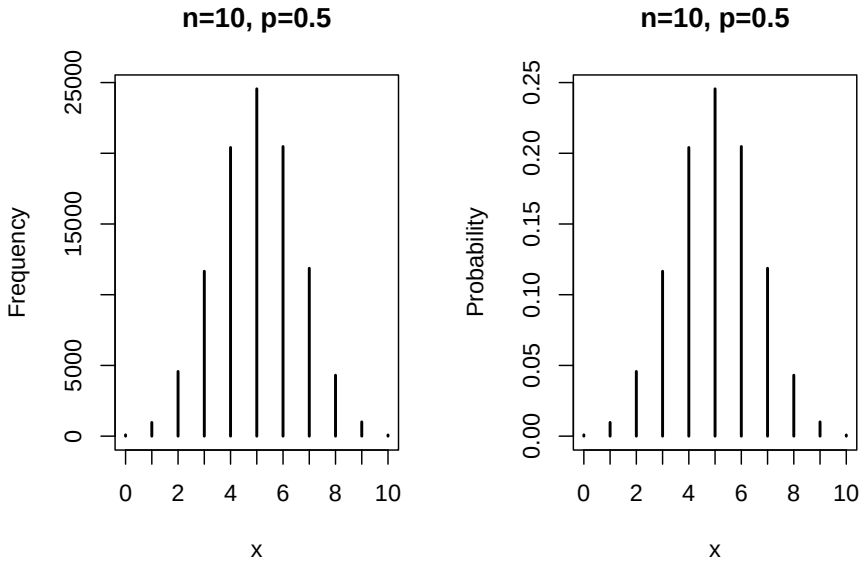


Figure 6.1: Flipping a coin ten times and repeating the experiment 1000 times.

6.2.1 Bernoulli Distribution

The Bernoulli (named after Jacob Bernoulli, 1654-1705) distribution is the simplest probability distribution. There are only two outcomes: *Success* and *Failure*. The distribution is characterized by the single parameter p. The probability mass function is written as $\Pr(X = 1) = p$ and, correspondingly, $\Pr(X = 0) = 1 - p$. The expected value and variance of the binomial is $E(x) = p$ and $Var(x) = p \cdot (1 - p)$. The distribution applied to many instances such as getting heads after one coin flip, getting a five after rolling a die, or voting for a particular candidate or party.

6.2.2 Geometric Distribution

The geometric distribution models the number of trials until the first success in a series of Bernoulli trials. Each trial has either "success" (with probability p) or "failure" (with probability $1-p$) as the two outcomes. The trials are independent and thus, probability of success remains constant across trials. The distribution can be expressed as follows:

$$\Pr(X = k) = (1 - p)^{k-1} \cdot p$$

where X is the number of trials until the first success, p is the probability of success on each trial, and k is the trial number where the first success occurs.

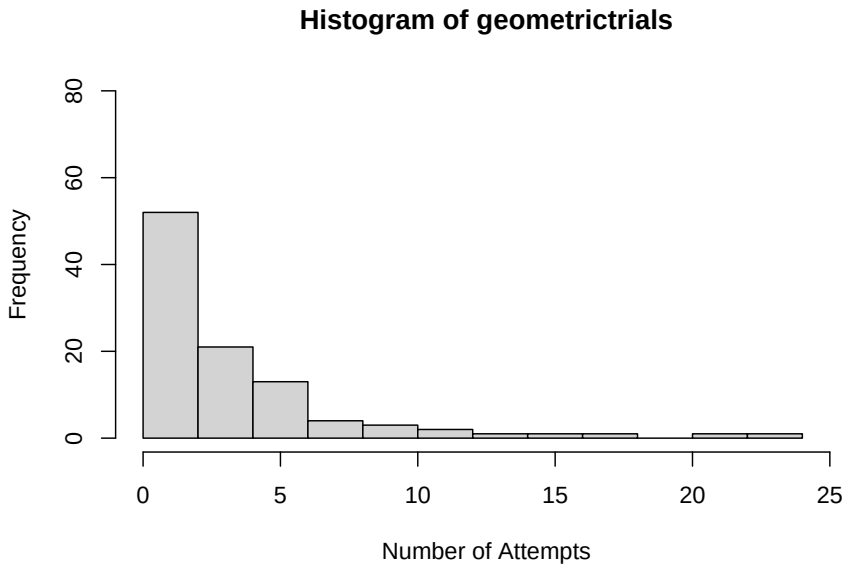


Figure 6.2: Example of a geometric distribution with a 20% chance of success at each trial simulated 100 times.

Assume $p = 0.2$, the we have the following:

$$\Pr(X = 3) = (1 - 0.2)^{3-1} \cdot 0.2 = (0.8)^2 \cdot 0.2 = 0.128$$

Thus, there is a 12.8% chance the first success occurs on the third trial. Expected value (i.e., average number of trials needed to get the first success) and variance are $E(X) = 1/p$ and $Var(X) = (1 - p)/p^2$, respectively. The geometric distribution can also be implemented in R. For example, assume that $p = 0.2$ leading to an expected value of $E(X) = 1/0.2 = 5$. That is, we need five trials on average for the first success.

```
geometrictrials = rgeom(100,prob=0.2)
hist(geometrictrials,xlab="Number of Attempts",breaks=10,
     xlim=c(0,25),ylim=c(0,80))
```

6.2.3 Binomial Distribution

Like the geometric distribution, the binomial distribution is closely related to Bernoulli because it represents repeated Bernoulli outcomes. The two parameters are n and p with the number of successes represented by k. The probability mass function is written as

$$\Pr(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

The expected value of X is $E(X) = n \cdot p$ and the variance is $Var(X) = n \cdot p \cdot (1-p)$. A situation must meet the following conditions for a random variable X to have a binomial distribution:

- You have a fixed number of trials involving a random process; let n be the number of trials.
- You can classify the outcome of each trial into one of two groups: success or failure.
- The probability of success is the same for each trial. Let p be the probability of success, which means $1 - p$ is the probability of failure.
- The trials are independent, meaning the outcome of one trial does not influence the outcome of any other trial.

Imagine a multiple choice test with three questions and five choices for each question. The binomial distribution is used to calculate the probability of two or more correct responses:

$$P(X = 2) = \binom{3}{2} \cdot 0.2^2 \cdot (1 - 0.2)^{3-2} = 0.096$$

and

$$P(X = 3) = \binom{3}{3} \cdot 0.2^3 \cdot (1 - 0.2)^{3-3} = 0.008$$

Summing up both probabilities leads to $\Pr(X \geq 2) = 0.104$. Let us do some additional examples assuming ten independent facility inspections with each facility having a probability of 10% of failing the inspection. The probability that exactly two fail can be represented as follows in R:

```
dbinom(2,size=10,prob=0.10)
```

```
## [1] 0.1937102
```

The probability that at most two fail can be written as

```
pbinom(2, size = 10, prob = 0.10)
```

```
## [1] 0.9298092
```

The probability that more than two fail can be expressed in two ways:

```
1-pbinom(2,size=10,prob=0.10)
```

```
## [1] 0.07019083
```

```
pbinom(2,size=10,prob=0.10,lower.tail=FALSE)
```

```
## [1] 0.07019083
```

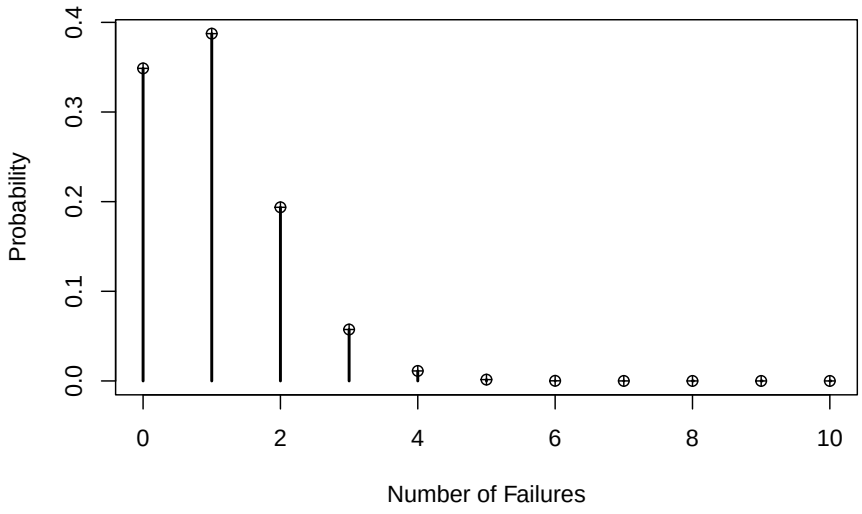
The probability that between two and five fail is written as:

```
sum(dbinom(2:5,size=10,prob=0.10))
```

```
## [1] 0.2637542
```

Great care must be exercised while using the option `lower.tail` in R. The Binomial distribution can also be visualized as follows.

```
x    = 0:10
prob = dbinom(x,size=10,prob=0.10)
plot(x,prob,type="h",lwd=2,
     xlab="Number of Failures",ylab="Probability")
points(x,prob,pch=10)
```



We can also simulate the outcomes associated with a Binomial distribution:

```
sim = rbinom(10000,size=10,prob=0.10)
mean(sim==2)
```

```
## [1] 0.1967
```

```
mean(sim<=2)
```

```
## [1] 0.9314
```

```
mean(sim>2)
```

```
## [1] 0.0686
```

The simulated probabilities will not be exactly equal to the theoretical probabilities since random number generation contains noise. As the number of simulations increases, the estimates generally get closer.

The binomial distribution can be used to analyze the issue of overbooking while highlighting that expected outcomes are not the same as outcomes at the expected value. Assume that an airline as a plane with a seating capacity of 100. The ticket price for each traveler is \$400. The airline can overbook the flight, i.e., selling more than 100 tickets, but has to pay \$700 in case a person has a valid ticket but needs to be re-booked to another flight. There is a probability of 10% that a booked passenger does not show up. If X is the number of passengers who show up, then

$$X \sim \text{Binomial}(n, 0.90)$$

where n is the number of tickets sold. A common mistake is to calculate the expected number who show up, $E(X) = 0.9 \cdot n$, plug that number into a profit equation, and call the result expected profit. In general, $E[g(X)] \neq g(E[X])$. To calculate expected profit correctly, we average profit across all possible numbers of passengers who may show up, weighted by their probabilities.

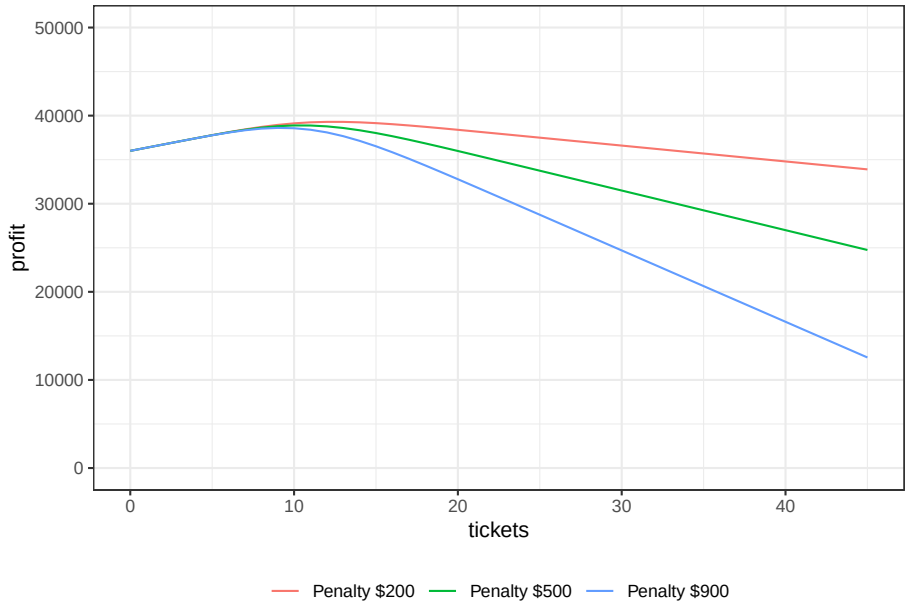


Figure 6.3: Example of using the binomial distribution to determine how many seats to sell over capacity.

6.2.4 Poisson Distribution

The Poisson distribution is used for count data (i.e., $0, 1, 2, \ldots$) within a fixed interval of time or space. For example, it can be used to model the number of calls within an hour or the number of defects per kilometer of pipe. Let $X \sim \text{Poisson}(\lambda)$ where λ is the expected number of events in the interval. That

is, $E(X) = \lambda$ but we also have $Var(X) = \lambda$. The probability mass function for the Poisson distribution is given by the following equation:

$$\Pr(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

An example of the Poisson distribution (named after Simeon Denis Poisson, 1781–1840) for different parameter values in Figure 6.4.

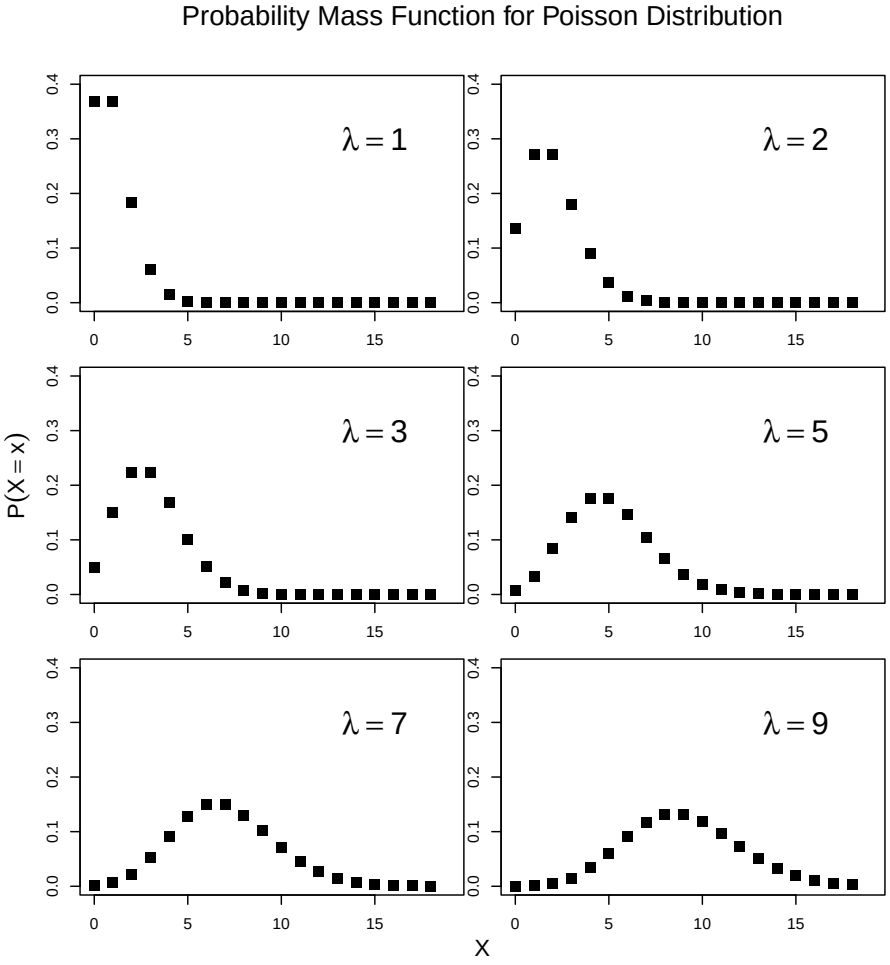


Figure 6.4: Poisson Distribution

Suppose a service center receives an average of 8 calls per hour.

```
lambda = 8
```

The probability of exactly 10 calls is

```
dpois(10, lambda = lambda)
```

```
## [1] 0.09926153
```

The probability of 10 or fewer calls is

```
ppois(10, lambda = lambda)
```

```
## [1] 0.8158858
```

The probability of more than 10 calls is

```
1 - ppois(10, lambda = lambda)
```

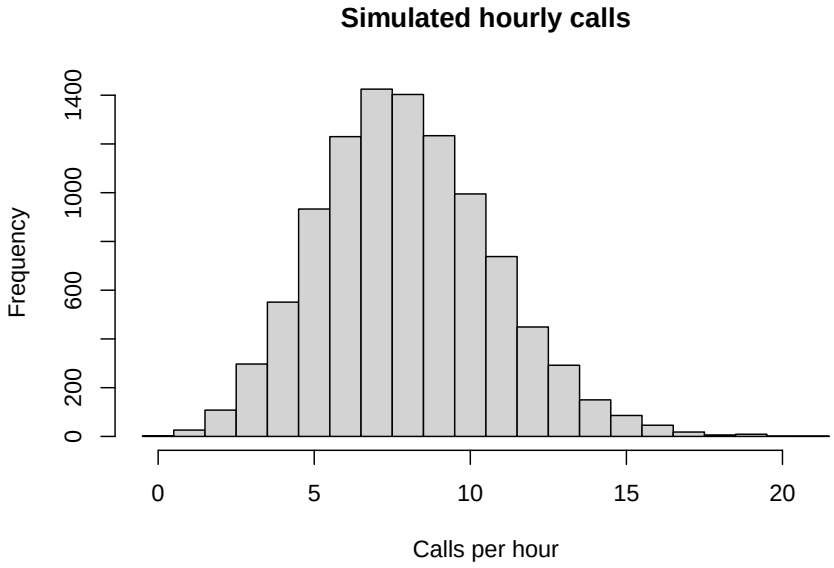
```
## [1] 0.1841142
```

We can also simulate entire hours:

```
set.seed(123)
sim_calls <- rpois(10000, lambda = 8)
mean(sim_calls)
```

```
## [1] 7.9691
```

```
hist(sim_calls,breaks=seq(-0.5,max(sim_calls)+0.5,by=1),
     xlab="Calls per hour",main="Simulated hourly calls")
```



A Poisson model is not appropriate simply because the variable is a count. For example, calls may arrive in clusters after a storm, accident, or system outage.

The average rate may differ dramatically between morning and evening. In those cases, the simple Poisson model may underestimate the amount of variation. The question is therefore not whether a variable is a count but whether the Poisson process a reasonable approximation to the way these counts are generated.

6.3 Continuous distributions

We define a random variable to be continuous if $F_X(x)$ is a continuous function of x and for every real number x. The probability density function, $f_X(x)$, of a continuous random variable X is the function $f(\cdot)$ that satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(u)du$$

Properties are that $f_X(x) \geq 0$ for all x and

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

It is important to note that for a continuous distribution, the probability of a particular event is zero. For a continuous random variable, $\Pr(X = x) = 0$. Probabilities are associated with intervals, not individual points. For example, $\Pr(60 < X < 70)$ is an area under the density curve.

6.3.1 Uniform distribution

The simplest continuous distribution is the uniform distribution. The distribution assumes that all values within an interval are equally likely. If $a < b$, a random variable X is said to have a uniform probability distribution on the interval (a, b) if and only if the density function of X is

$$f(x) = \frac{1}{b-a}$$

Suppose that a package is scheduled to be delivered between 9:00 (9 AM) and 20:00 (8 PM). You are out for lunch between 12:00 (noon) and 13:30 (1:30 PM). The probability that the package arrives during the lunch break is $\Pr(12 < x < 13.5) = \frac{1}{20-9} \cdot 1.5 = 13.64\%$, which is equivalent to 1.5/11. This can be calculated in R as well:

```
punif(13.5,min=9,max=20)-punif(12,min=9,max=20)
```

```
## [1] 0.1363636
```

6.3.2 Normal distribution

The Normal distribution is one of the most frequently used models in statistics. The random variable X is said to be normally distributed with mean μ and

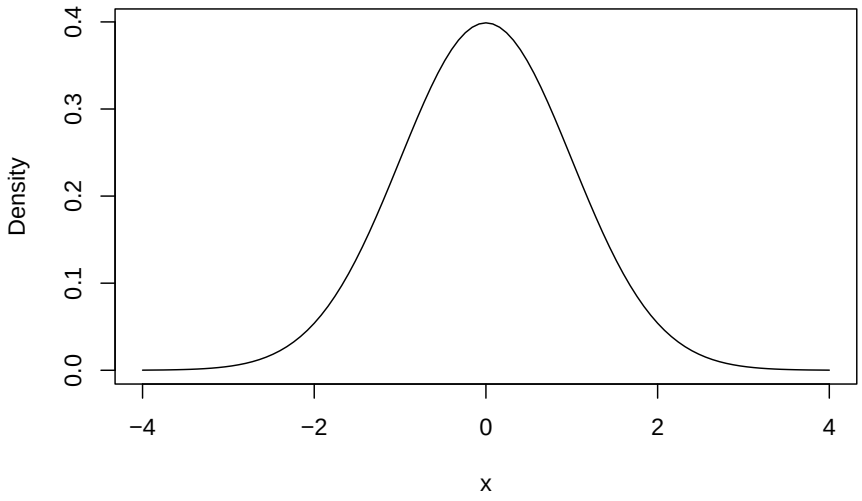


Figure 6.5: Standard Normal distribution with N(0,1)

variance σ^2 (abbreviated by $x \sim N[\mu, \sigma^2)$ if the density function of x is given by

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

The Normal probability density function is bell-shaped and symmetric (Figure 6.5. The next chapter on sampling demonstrates that the Normal Distribution holds a particular place among all distributions due to its relation to the so-called Central Limit Theorem. The Normal distribution can be derived from the Binomial Distribution as shown in Figure 6.6. The experiment associated with deriving the Normal distribution from the bionomial distribution is usually done with a so-called Galton Board. For more information on how the Normal Distribution evolved, see The Evolution of the Normal Distribution.

Note that any random variable following a Normal distrubtion with mean μ and variance σ^2 can be expressed as a Standard Normal distrubtion with $\mu = 0$ and $\sigma = 1$ by calculating the z-value:

$$z = \frac{X - \mu}{\sigma}$$

where z is the distance from the mean of a Normal distribution expressed in units of the standard deviation.

Suppose that the starting salary after graduate school is approximately normally distributed with mean \$75,000 and standard deviation \$10,000 (in what follows,

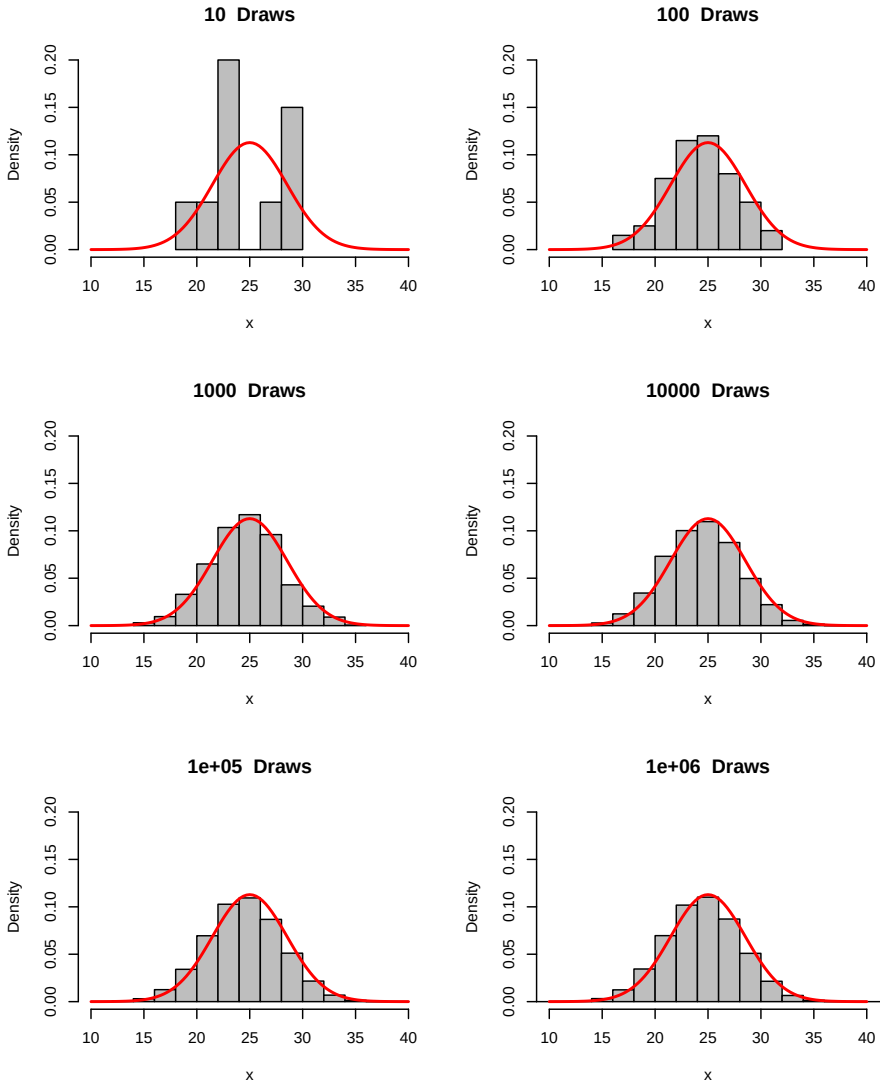


Figure 6.6: Derivation of the Normal Distribution from the Binomial Distribution

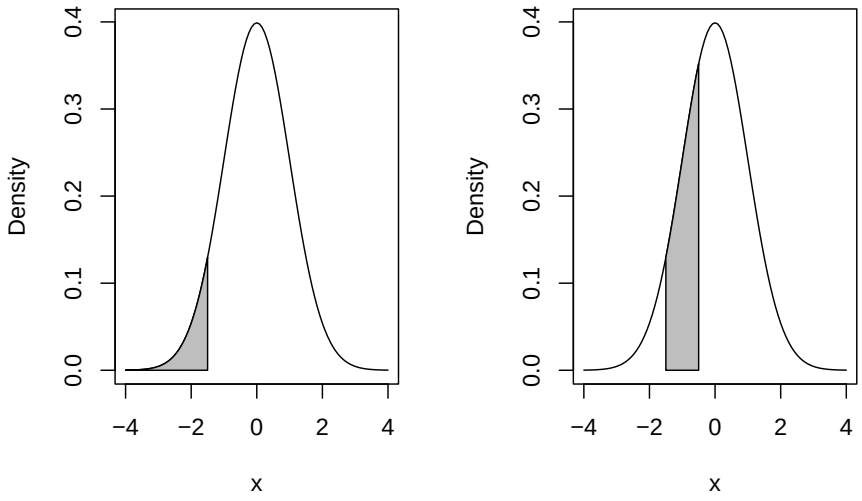


Figure 6.7: The left panel shows the probability below a z-value of -1.5 and the right panel shows the probability of z values between -1.5 and 0.5.

let us express this in \$1,000). Thus, we have $\mu = 75$ and $\sigma = 10$ and we want to find $P(60 < x < 70)$. We can do this by calculating the z values for 60 and 70 as follows:

$$\Pr(x < 60) \Rightarrow z_1 = \frac{60 - 75}{10} = -1.5 P(x < 70) \Rightarrow z_2 = \frac{70 - 75}{10} = -0.5$$

So 60 and 70 are 1.5 and 0.5 standard deviations below the mean, respectively. Historically, standardization was essential because probabilities were looked up in printed Normal tables. R can calculate Normal probabilities directly, so you do not need to convert every problem to a z-score first. The z-score remains useful for interpretation.

```
pnorm(70,mean=75,sd=10)-pnorm(60,mean=75,sd=10)
```

```
## [1] 0.2417303
```

What salary marks the lowest 5% of cases? We need the 95th percentile:

```
qnorm(0.05,mean=75,sd=10)
```

```
## [1] 58.55146
```

6.3.3 *t*-Distribution

The Student's t-distribution is a probability distribution that is similar in shape to the Normal distribution but has heavier tails. It is commonly used when estimating a population mean when the population standard deviation is unknown and must be estimated from the sample. The shape of the t-distribution depends on its degrees of freedom, which are typically related to the sample size. With small samples, the heavier tails reflect the additional uncertainty from estimating the standard deviation. As the sample size and degrees of freedom increase, the t-distribution approaches the standard Normal distribution. This distribution was published by William Gosset in 1908. His employer, Guinness Breweries, required him to publish under a pseudonym, so he chose "Student".

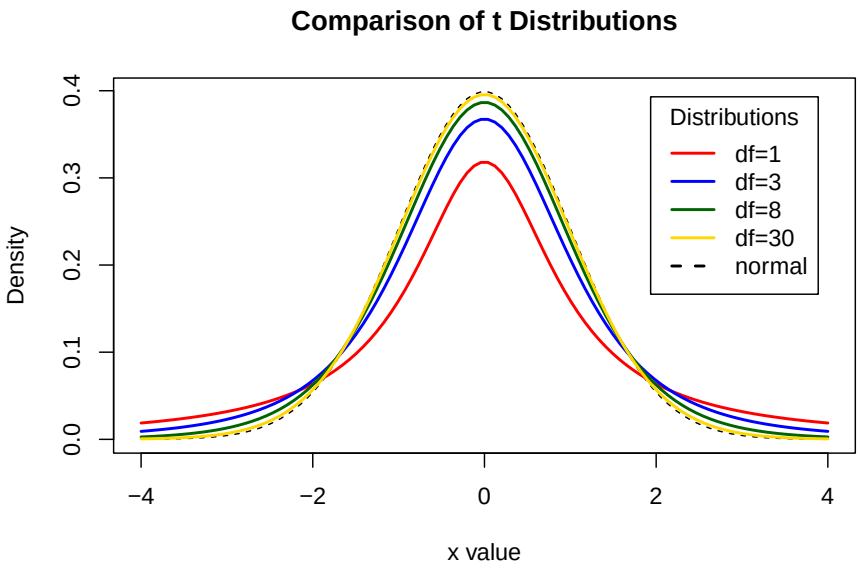


Figure 6.8: Student Distribution for various levels of degrees of freedom and Normal distribution

6.3.4 Exponential Distribution

The Exponential distribution is useful for positive waiting times when events occur according to a constant-rate Poisson process. If events occur at rate λ, then the waiting time T until the next event can be modeled as follows with expected value $E(T) = 1/\lambda$:

$$T \sim \text{Exponential}(\lambda)$$

Suppose customers arrive at a desk at an average rate of 0.5 per minute. Under

a simple Poisson-process model, the waiting time until the next arrival is Exponential with $\lambda = 0.5$. The average waiting time is therefore $1/0.5 = 2$ minutes. The probability of waiting more than 3 minutes is

```
1-pexp(3,rate=0.5)
```

```
## [1] 0.2231302
```

The probability of waiting between 1 and 4 minutes is

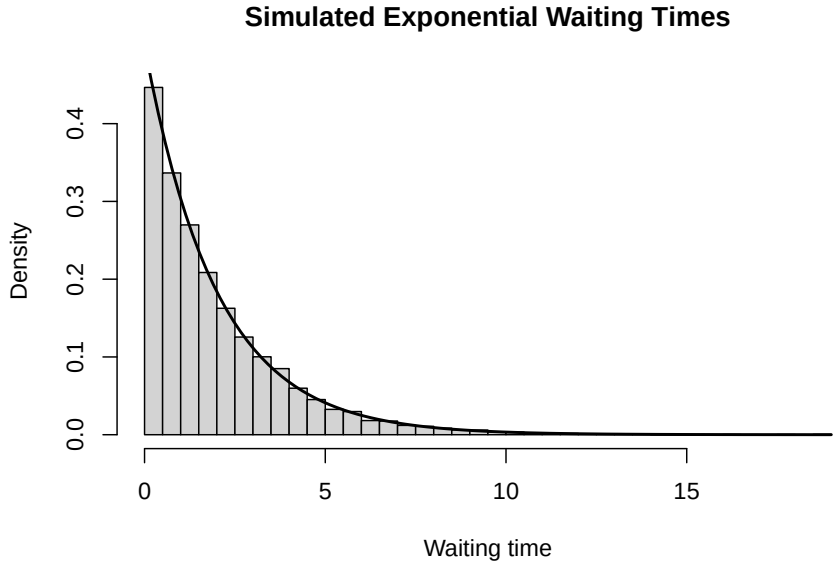
```
pexp(4,rate=0.5)-pexp(1,rate=0.5)
```

```
## [1] 0.4711954
```

Simulated waiting times can be generated with

```
wait = rexp(10000,rate=0.5)
hist(wait,breaks=40,probability=TRUE,
     xlab="Waiting time",
     main="Simulated Exponential Waiting Times")

curve(dexp(x, rate = 0.5),
      add = TRUE,
      lwd = 2)
```



The Exponential distribution has a strong assumption: the event rate does not depend on how long we have already waited. This is called the memoryless property. That assumption is plausible for some processes and implausible for

others.

6.4 Exercises

1. ***Distribution Choice*** (***): For each situation below, choose the most reasonable model from: (1) Binomial, (2) Poisson, (3) Normal, (4) Exponential, and (5) Uniform. For each answer, write one sentence about the process, not merely the name of the distribution. Then give all five situations to AI and ask to classify them. Select two cases where its justification is weakest or where multiple answers could be defensible. Explain your reasoning.

    - Number of pothole complaints received by a city during one day.
    - Number of defective items in a sample of 50 independently selected products when the defect probability is constant.
    - Time until the next 911 call under a constant-rate Poisson process.
    - Adult heights within a relatively homogeneous population.
    - Household income in a large metropolitan area.

2. ***Bus Delays*** (***): A regional transit agency reports that a bus on a particular route has a 12% probability of arriving more than 10 minutes late. Assume delays for different trips are independent and the probability remains constant. Consider the next 12 trips.

    a. Define the random variable and identify an appropriate probability distribution. What is the expected number of late trips?
    b. Calculate the following probabilities: (1) Exactly 2 trips are more than 10 minutes late, (2) at most 2 trips are more than 10 minutes late, and (3) more than 2 trips are more than 10 minutes late.
    c. Simulate 100,000 sets of 12 trips and estimate the same three probabilities.
    d. Ask an AI tool to solve part (b). Compare its answer with your R calculations from part (c). Identify any difference in notation, logic, rounding, or interpretation. If there is no numerical mistake, identify one assumption that the AI should have discussed more explicitly.

3. ***Apple Juice Machine*** (**): An apple juice company wants to purchase a new bottling machines to fill 16 ounce cans. Two manufactures indicate the following performances for their machines. The first machine fills the cans with 16.5 ounces and a standard deviation of 0.3. The second machine fills the cans with 16.2 ounces and standard deviation 0.1. The filling quantities for both machines are normally distributed. Anything below 16 ounces cannot be sold since it does not meet the advertised 16 ounces.

    a. For each machine, calculate the probability that a can contains less than 16 ounces.

   b. Out of 100,000 cans, calculate the expected number of cans that cannot be sold for each machine.
   c. Simulate 100,000 cans from each machine and compare the simulated rate of cans containing less than 16 ouces to the theoretical probability.
   d. If the only objective is to minimize the number of cans below 16 ounces, which machine performs better?
   e. Ask AI which machine the company should buy. Discuss why minimizing the number of cans filled below 16 ounces may be an incomplete business objective. What additional information would be necessary for a more complete decision?

4. ***City Service Center*** (***): A city service center receives an average of 9.5 calls every 30 minutes. Assume a Poisson model is reasonable.

   a. Calculate the probability of exactly 10 calls during a 30-minute period.
   b. Calculate the probability of more than 15 calls.
   c. Calculate the smallest call-center capacity such that the center can handle the number of incoming calls in at least 95% of 30-minute periods without a queue forming. Note that this is a question relating to quantiles.
   d. Simulate 50,000 30-minute periods and estimate the proportion in which calls exceed the capacity found in part (c).
   e. Give two real-world reasons why a constant-rate Poisson model might fail for service-center calls.

5. ***Customer Service*** (***): Suppose customers arrive at a public service desk at an average rate of one customer every 4 minutes. Assume arrivals follow a constant-rate Poisson process.

   a. What Exponential rate parameter corresponds to this process when time is measured in minutes?
   b. Calculate the probability that the next customer arrives within 2 minutes.
   c. Calculate the probability that the wait exceeds 10 minutes.
   d. Simulate 20,000 waiting times and compare the simulation with your theoretical answers.
   e. The service desk closes for lunch and experiences a rush immediately after reopening. Explain why the Exponential model with one constant rate may then be inappropriate.
   f. Ask AI to explain the memoryless property of the Exponential distribution using this example. Rewrite the explanation in two sentences that you believe a city manager with no statistics background would understand.

6. ***Fair Game*** (***): A friend of yours has a coin and proposes the following game. You toss the coin 10 times and count the number of heads. The amount b you win or lose on k heads is given by $b = 4 \cdot \sqrt{k} - 2 \cdot k$.

   a. Plot the payoff function with the number of heads on the horizontal

axis and payoff in dollars on the vertical axis.
b. Make an exact computation using R to decide if this is a good bet.
c. Use the function `rbinom()` and generate 100,000 random outcomes of the game. Calculated the expected winning and compare it to your calculation in part (b). Include a histogram of the random draws.
d. Change the payoff to $b = \alpha \cdot \sqrt{k} - 2 \cdot k$. Use AI to find approximately how large α must be before the expected payoff becomes positive.

7. ***Overbooking*** (***): An aircraft has 115 seats. Each ticketed passenger independently shows up with probability 0.90. The airline earns $400 for each passenger who travels. A passenger who shows up but cannot be seated costs the airline $700 in compensation.

   a. Write a function that calculates profit for a given number of passengers who show up.
   b. Write a second function that calculates expected profit for a given number of tickets sold by averaging profit across all possible numbers of passengers who may show up.
   c. Calculate expected profit when 115 through 145 tickets are sold.
   d. Plot expected profit against the number of tickets sold.
   e. Repeat the analysis for compensation costs of $300, $700, and $1,500. Explain how the penalty changes the decision.
   f. Ask AI to explain why calculating profit using only the expected number of passengers who show up is generally incorrect. Verify the explanation numerically for at least one ticket-sales level by comparing $E[g(X)]$ to $g(E[X])$.

8. ***Public Affairs Problem Creation*** (***): Create your own probability problem related to public affairs, policy, management, or a topic that interests you. Your problem must use one of the distributions from this chapter. Describe the real-world process and define the random variable. State the distribution and its parameter values. Ask one probability question and one quantile or expected-value question related to the created problem. Solve both questions in R and verify at least one answer using simulation with at least 50,000 draws. Next, ask AI to solve your problem and evaluate its solution. Your AI evaluation must answer the following: Did it select the same model? Did it use the same parameters? Did it translate the probability statement correctly? Did its numerical answer agree with R? What assumption is most important for determining whether the model is reasonable?

9. ***O'Neill Air*** (**): The probability of any O'Neill Air flight being delayed more than 15 minutes is 0.1. We randomly select four different O'Neill Air flights.

   (a) Calculate the probability that all four flights arrived within 15 minutes of the scheduled time?
   (b) Calculate the probability that none of the selected flights arrived

within 15 minutes of the scheduled time?

(c) Calculate the probability that at least one of the selected flights arrived within 15 minutes of the scheduled time?

10. ***ACT*** (**): Assume ACT scores are normally distributes with mean 19 and standard deviation of 4. A university accepts applicants which score in the top 20%. What is the minimum ACT score that gets you accepted? Calculate the probability of 0, 1, ..., 10 students meetings the requirements out of 10 students.

11. ***LED Light Bulbs*** (**): On average, a LED light bulb manufacturer produces 250 defective light bulbs out of 5,000. You randomly pick 8 light bulbs. What is the probability of finding more than 1 defective bulbs?

12. ***Vaccine Trials*** (**): A pharmaceutical company estimates that a new vaccine that is undergoing test trials has a 1/10000 chance of causing a serious side effect in a human being. If the company administers the vaccine to 10,000 volunteers, what is the probability that at least 1 volunteers develops those serious side effects? The company wants to make sure that the probability of at least one person developing the side effect is 95%. How many volunteers do they need?

13. ***Binomial Probabilities versus Simulation*** (***): A coin has probability $p = 0.6$ of landing heads. Let $X \sim \text{Binomial}(n = 50, p = 0.6)$ be the number of heads in 50 flips.

    a. Compute the exact probabilities for $\Pr(X = 30)$ and $\Pr(25 \leq X \leq 35)$.
    b. Calculate $E[X]$ and $Var(X)$
    c. Simulate 10,000 independent experiments of 50 flips each and estimate $(\Pr(25 \leq X \leq 35))$ as a relative frequency.
    d. Make a histogram of the simulated (X) values and add vertical lines at 25 and 35. Briefly compare the exact probability from (a) to your simulation estimate.

14. ***Waiting Times and CLT for Sample Means*** (***): Suppose waiting time (in minutes) for a service is Exponential with rate $(\lambda = 0.5)$. Thus $(T \sim \text{Exp}(\lambda = 0.5))$.

    a. Compute the theoretical values of (a) $(\mathbb{E}[T])and(\text{Var}(T))$, (b) $(\Pr(T > 3))$, and (c) $\Pr(1 < T < 4)$.
    b. Simulate 10,000 independent draws from (Exp(0.5)). Estimate the probabilities in (1b)–(1c) by relative frequencies and compare to the exact values.
    c. Let $(\bar{T} * 30)$ be the mean of (k=30) i.i.d. Exponential(0.5) draws. Use the Central Limit Theorem to approximate the distribution of $(\bar{T} * 30)$. Simulate 5,000 sample means $(\bar{T}_{30})$.

    - Plot a histogram of the 5,000 means.

- On a new plot, overlay the Normal density with mean ($\mu = \mathbb{E}[T]$) and standard deviation ($\sigma_{\bar{T}} = \sqrt{\mathrm{Var}(T)/30}$). (You can plot the theoretical curve by evaluating `dnorm` on a grid and using `lines`.)